

The main goal of this section is to prove the following fact:

Theorem 16.1

If G is a finite abelian group then G is isomorphic to a direct product of cyclic groups whose orders are powers of primes:

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \dots \times \mathbb{Z}_{p_k^{r_k}}$$

for primes $p_1, \dots, p_k$ and integers $r_1, \dots, r_k \geq 1$ such that $p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_k^{r_k} = |G|$.

Example. Take $72 = 2^3 \cdot 3^2$. Theorem 16.1 says that every abelian group of order 72 is isomorphic to one of the following groups:

$$\begin{array}{ll} \mathbb{Z}_8 \times \mathbb{Z}_9 & \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \end{array}$$

Definition 16.2

A *short exact sequence* of groups is a sequence group homomorphisms

$$K \xrightarrow{i} G \xrightarrow{q} H$$

such that:

- i is 1-1
- q is onto
- $\text{Im}(i) = \text{Ker}(q)$

Example. Let K be a normal subgroup of G and let $i: K \rightarrow G$ be the inclusion homomorphism: $i(a) = a$. Also, let $q: G \rightarrow G/K$ be the quotient homomorphism $q(a) = aK$. This defines a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} G/K$$

Example. If $f: G \rightarrow H$ is homomorphism which is onto, then we have a short exact sequence

$$\text{Ker}(f) \xrightarrow{i} G \xrightarrow{f} H$$

where $i: \text{Ker}(f) \rightarrow G$ is the inclusion homomorphism.

Example. Let G, H be groups. Define $i: G \rightarrow G \times H$ by $i(g) = (g, e)$, and $q: G \times H \rightarrow H$ by $q(g, h) = h$. This defines a short exact sequence

$$G \xrightarrow{i} G \times H \xrightarrow{q} H$$

Notice that in this case we also have a homomorphism $s: H \rightarrow H \times G$, $s(h) = (e, h)$ and $q \circ s = \text{id}_H$.

Theorem 16.3

Consider a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} H$$

where K, G, H are abelian groups. Assume that there exists a homomorphism $s: H \rightarrow G$ such that $q \circ s(h) = h$ for all $h \in H$. Then $G \cong K \times H$.

Note. Theorem 16.3 is not true in general for non-abelian groups. For example, in the dihedral group D_4 take the subgroup of rotations $K = \{I, R_{90}, R_{180}, R_{270}\}$. Since K is a normal subgroup of D_4 , this gives a short exact sequence

$$K \xrightarrow{i} D_4 \xrightarrow{q} D_4/K$$

The group D_4/K consists of two cosets: IK and VK . Define $s: D_4/K \rightarrow D_4$ by $s(IK) = I$ and $s(VK) = V$. One can check that this is a group homomorphism. Moreover, $q(s(IK)) = IK$ and $q(s(VK)) = VK$. However, D_4 is not isomorphic to $K \times D_4/K$. Indeed, since $K \cong \mathbb{Z}_4$ and $D_4/K \cong \mathbb{Z}_2$, thus $K \times D_4/K \cong \mathbb{Z}_4 \times \mathbb{Z}_2$ is an abelian group, while D_4 is non-abelian.

Proof of Theorem 16.3. Define a function $f: K \times H \rightarrow G$ by $f(k, h) = i(k) \cdot s(h)$. We will show that this function is an isomorphism of groups.

First, we check that f is a homomorphism:

$$\begin{aligned}
 f((k, h) \cdot (k', h')) &= f(kk', hh') \\
 &= i(kk') \cdot s(hh') \\
 &= i(k) \cdot i(k') \cdot s(h) \cdot s(h') \\
 &= (i(k) \cdot s(h)) \cdot (i(k') \cdot s(h')) \\
 &= f(k, h) \cdot f(k', h')
 \end{aligned}$$

Next, assume that $(k, h) \in \text{Ker}(f)$. Then $f(k, h) = i(k) \cdot s(h) = e$. This gives:

$$e = q(i(k) \cdot s(h)) = q(i(k)) \cdot q(s(h)) = e \cdot h = h$$

so $e = h$. Thus, $e = f(k, h) = f(k, e) = i(k) \cdot e = i(k)$. Since i is 1-1, we get that $k = e$. Therefore the only element in $\text{Ker}(f)$ is the identity element (e, e) , which means that f is 1-1.

It remains to show that f is onto. Take an element $g \in G$, and let $h = q(g)$. We have

$$q(gs(h)^{-1}) = q(g) \cdot q(s(q(g)^{-1})) = q(g) \cdot q(g^{-1}) = e$$

which shows that $gs(h)^{-1} \in \text{Ker}(q)$. By exactness, we have $\text{Ker}(q) = \text{Im}(i)$, so there is an element $k \in K$ such that $i(k) = g \cdot s(h)^{-1}$. Consider the element $(k, h) \in K \times H$. We have:

$$f(k, h) = i(k) \cdot s(h) = g \cdot s(h)^{-1} \cdot s(h) = g$$

□

Corollary 16.4

Consider a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} H$$

where K, G, H are abelian groups. Assume that there exists a homomorphism $s: H \rightarrow G$ such that $q \circ s$ is an isomorphism. Then $G \cong K \times H$.

Proof. Define $f = (q \circ s)^{-1} \circ q: G \rightarrow H$. The sequence

$$K \xrightarrow{i} G \xrightarrow{f} H$$

is a short exact sequence. Moreover, we have $f \circ s = (q \circ s)^{-1} \circ q \circ s = \text{id}_H$. Thus by Theorem 16.3 we get $G \cong K \times H$. □

Theorem 16.5

Let G be a finite abelian group. Assume that $|G| = p^r m$ where p is a prime, $r \geq 1$ and m is a number which is not divisible by p . Then $G = K \times H$ where $|K| = p^r$ and $|H| = m$.

Lemma 16.6

Let G be a finite abelian group. Assume that there exists a prime p such that the order of each element $g \in G$ is a power of p . For $m \in \mathbb{Z}$ consider the function

$$f: G \rightarrow G$$

given by $f(g) = g^m$. If m is not divisible by p then f is a group isomorphism.

Proof. By Corollary 15.8 we have $|G| = p^r$ for some $r \geq 0$. Therefore $g^{p^r} = e$ for all $g \in G$.

Since $\gcd(m, p^r) = 1$, there exist $k, l \in \mathbb{Z}$ such that $km + lp^r = 1$. Define $s: G \rightarrow G$ by $s(g) = g^k$. We have

$$s \circ f(g) = (g^m)^k = g^{km} = g^{1-lp^r} = g \cdot g^{-lp^r} = ge = g$$

so $s \circ f = \text{id}_G$. Similarly, $f \circ s = \text{id}_G$. Thus f is an isomorphism and $f^{-1} = s$. $\square$

Proof of Theorem 16.5. Define

$$H := \{g \in G \mid |g| = p^i \text{ for some } i \geq 1\}$$

One can check that this is a subgroup of G . Notice that if $g \in G$ then $g^m \in H$. Indeed, we have

$$(g^m)^{p^r} = g^{mp^r} = g^{|G|} = e$$

so $|g^m|$ divides p^r . As a consequence we obtain a homomorphism $q: G \rightarrow H$, $q(g) = g^m$. This homomorphism is onto since, by Lemma 16.6, $q|_H: H \rightarrow H$ is an isomorphism. Take the short exact sequence

$$\text{Ker}(q) \longrightarrow G \xrightarrow{q} H$$

Define $s: H \rightarrow G$ by $s(h) = h$. The composition $q \circ s: H \rightarrow H$ is given by $q \circ s(h) = h^m$ which is an isomorphism by Lemma 16.6. Using Corollary 16.4 we obtain that $G \cong \text{Ker}(q) \times H$.

We have $|H| \cdot |\text{Ker}(q)| = |G| = p^r m$. By Corollary 15.8, $|H|$ is a power of p . Also, since $\text{Ker}(q) = \{g \in G \mid g^m = 1\}$, thus the order of every element of $\text{Ker}(q)$ divides m . This implies that $\text{Ker}(q)$ does not contain any elements of order p . By Cauchy Theorem 15.6, we obtain that $|\text{Ker}(q)|$ is not divisible by p . Therefore $|H| = p^r$ and $|\text{Ker}(q)| = m$. $\square$

Corollary 16.7

If G is a finite abelian group and $|G| = p_1^{r_1} p_2^{r_2} \cdot \dots \cdot p_k^{r_k}$ where $p_1, p_2, \dots, p_k$ are distinct primes then

$$G = G_1 \times G_2 \times \dots \times G_k$$

where $|G_i| = p_i^{r_i}$.

Proof. We use induction with respect to the number of distinct primes k . If $k = 1$ then $|G| = p_1^{r_1}$, so we take $G_1 = G$.

Assume that the statement is true for all abelian groups whose order is a product of powers of k distinct primes, and let G be a group such that

$$|G| = p_1^{r_1} p_2^{r_2} \cdot \dots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$$

where $p_1, \dots, p_{k+1}$ are distinct primes. By Theorem 16.5 we get $G = G_1 \times H$ where $|G_1| = p_1^{r_1}$ and $|H| = p_2^{r_2} \cdot \dots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$. By the inductive assumption, $H \cong G_2 \times \dots \times G_{k+1}$ where $|G_i| = p_i^{r_i}$. This gives

$$G \cong G_1 \times H \cong G_1 \times G_2 \times \dots \times G_{k+1}$$

$\square$

Theorem 16.8

If G is an abelian group such that $|G| = p^n$ for some prime p then G is a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p^{k_1}} \times \mathbb{Z}_{p^{k_2}} \times \dots \times \mathbb{Z}_{p^{k_m}}$$

for some $k_1, k_2, \dots, k_m$.

Idea of Proof. Use induction with respect to the order of the group G . Let p^r be the largest order of an element in G , and let $a \in G$ be an element such that $|a| = p^r$.

One can show that there exists a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} \mathbb{Z}_{p^r}$$

such that $q(a) = 1 \in \mathbb{Z}_{p^r}$. Take the homomorphism $s: \mathbb{Z}_{p^r} \rightarrow G$ given by $s(k) = a^k$. Since $q \circ s(k) = k$ for all $k \in \mathbb{Z}_{p^r}$, by Theorem 16.3 we obtain that $G \cong \mathbb{Z}_{p^r} \times K$. By the inductive assumption we get that K is a direct product of cyclic groups, $K \cong \mathbb{Z}_{p^{k_1}} \times \dots \times \mathbb{Z}_{p^{k_m}}$, which gives $G \cong \mathbb{Z}_{p^r} \times (\mathbb{Z}_{p^{k_1}} \times \dots \times \mathbb{Z}_{p^{k_m}})$. $\square$

Proof of Theorem 16.1. It follows from Corollary 16.7 and Theorem 16.8. $\square$